



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

2003
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 3

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections. Section A (Questions 1 and 2), Section B (Questions 3 and 4) and Section C (Questions 5 and 6).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 84 Marks

- All questions are of equal value.

Examiner: *B. Opferkuch*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

QUESTION 1.Use a *separate* Writing Booklet**Marks**

- (a) Find a primitive of $\frac{1}{\sqrt{1-x^2}}$. 1
- (b) Evaluate $\int_0^{\ln 5} e^{-x} dx$ 2
- (c) Find $\lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{h}$ 1
- (d) A box contains 5 red marbles and 4 white marbles.
Three marbles are drawn in succession without replacement,
What is the probability that any two marbles are red and one is white? 2
- (e) The sector OAB of a circle centre O and radius r has an area of $\frac{3\pi}{4} \text{ cm}^2$. 2
If the arc AB subtends an angle of $\frac{\pi}{6}$ at O , find the length of the arc AB .
- (f) If $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \tan^{-1}(1) = k\pi$ (where k is rational) 2
Find the value of k .
- (g) Find $\int \frac{\ln 3x}{x} dx$, using the substitution $u = \ln 3x$. 4

QUESTION 2.Use a *separate* Writing Booklet.**Marks**(a) Differentiate the following with respect to x .**8**

(i) $x \cos x$

(ii) $\log_e(\cos 5x)$

(iii) $\tan^{-1}\left(\frac{x}{3}\right)$

(iv) $e^{\tan x}$

(v) $\left\{1 + \cos^{-1}(3x)\right\}^3$

(b) Find a primitive of:

6

(i) $x^2 e^{2x^3+1}$

(ii) $\frac{5}{4+x^2}$

(iii) $\frac{x^2}{x^3+7}$

(iv) $-\sin(\pi - x)$

QUESTION 3.Use a *separate* Writing Booklet.**Marks**

- (a) In choosing three letters from the word PROBING, and assuming each choice is equally likely, what is the probability of choosing just one vowel? **2**
- (b) (i) In how many ways can the numbers 1, 2, 3, 4, 5, 6 be arranged around a circle? **3**
- (ii) How many of these arrangements have at least two even numbers together?
- (c) (i) Show that the function $f(x) = x^2 + e^{-\frac{1}{2}x} - 5$ has a root between -2 and -1 . **4**
- (ii) Taking $x = -2$ as a first approximation, apply Newton's method once, to show that the root of $f(x) = 0$ is approximately $-\frac{18}{e+8}$.
- (d) (i) Show that $\frac{1+x}{1-x} = -1 + \frac{2}{1-x}$. **3**
- (ii) Hence find $\int \frac{1+x}{1-x} dx$
- (e) Write down the general solution of $\sqrt{3} \tan \theta - 1 = 0$. **2**
Leave answer in exact radian form.

QUESTION 4.Use a *separate* Writing Booklet**Marks**

(a) Consider the curves $y = \sin x$ and $y = \cos 2x$. **5**

(i) Sketch the graphs of the two curves on the same axes,
in the domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{6}$.

(ii) Show that the curves intersect at $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{6}$.

(iii) Hence find the exact area bounded by the two curves.

(b) Consider the function $f(x) = 3\sin^{-1}\left(\frac{x}{2}\right)$. **3**

(i) Evaluate $f(2)$.

(ii) State the domain and range of $y = f(x)$.

(iii) Sketch the graph of $y = f(x)$.

(c) Find the exact value of $\tan\left[2\tan^{-1}\left(-\frac{1}{2}\right)\right]$. **3**

(d) The function $g(x)$ is given by $g(x) = \cos^{-1} x + \sin^{-1} x$, $0 \leq x \leq 1$. **3**

(i) Find $g'(x)$.

(ii) Sketch the graph of $y = g(x)$.

QUESTION 5.Use a *separate* Writing Booklet**Marks**

(a) Consider the function $f(x) = x \sin^{-1}(x^2)$. **8**

- (i) State the domain and range of $f(x)$.
- (ii) Find $f'(x)$.
- (iii) Show that there is a horizontal inflexion at the origin.
- (iv) What is the slope of the tangent at $x = 1$.
- (v) Sketch the curve which represents $f(x)$.

(b) (i) Show that $\int \frac{x}{\sqrt{x^2 + 2x - 3}} dx = \int \frac{x}{\sqrt{(x+1)^2 - 4}} dx$. **3**

(ii) Using the substitution $2 \sec u = x + 1$, find $\int \frac{x}{\sqrt{x^2 + 2x - 3}} dx$

(c) The curve $y = \frac{1}{\sqrt{1+x^2}}$ is rotated about the x -axis. **3**

Find the volume of the solid enclosed between $x = \frac{1}{\sqrt{3}}$ and $x = \sqrt{3}$.

QUESTION 6.Use a *separate* Writing Booklet**Marks**

(a) Evaluate $\int_0^{\sqrt{2}} x \sqrt[3]{x^2 + 1} \, dx$ – using the substitution $u = x^2 + 1$.

3

Answer in exact form.

(b) Consider the equation $\cos 3x = \sin x$

3

(i) Show that $\cos\left(\frac{\pi}{2} - A\right) = \sin A$

(ii) Hence, or otherwise, find the general solution of the equation $\cos 3x = \sin x$.

(c) (i) Show that $\frac{d}{dx} \tan^3 x = 3 \sec^4 x - 3 \sec^2 x$.

4

(ii) Using (i) or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$ –

(d) Show that the exact value of $\int_0^{\frac{\pi}{6}} \sin^2 x \, dx = \frac{2\pi - 3\sqrt{3}}{24}$

4

THIS IS THE END OF THE PAPER

(1) (a) $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}x + c$

(b) $\int_0^{\ln 5} e^{-x} dx = -e^{-x} \Big|_0^{\ln 5}$
 $= -[e^{-\ln 5} - e^0]$
 $= -[e^{\ln \frac{1}{5}} - 1]$
 $= 1 - \frac{1}{5} = \frac{4}{5}$

$e^{\ln x} = x$

(c) $\lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} = \frac{1}{2} \lim_{h \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}}$
 $= \frac{1}{2} \times 1$
 $= \frac{1}{2}$

$\left[\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1 \right]$

(d) 5R 4W

method 1:

$\frac{\binom{5}{2} \times \binom{4}{1}}{\binom{9}{3}} = \frac{10}{21}$

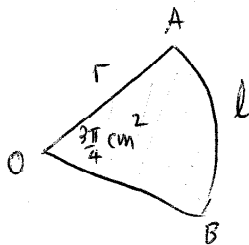
method 2: RRW

$\left(\frac{5}{9} \times \frac{4}{8} \times \frac{4}{7} \right)$

x3

$= \frac{10}{21}$

(e)



$\therefore \frac{1}{2} r^2 \theta = \frac{3\pi}{4}$

$\therefore \frac{1}{2} r^2 \times \frac{\pi}{6} = \frac{3\pi}{4}$

$\therefore r^2 = \frac{3}{4} \times 12 = 9$

$\therefore r = 3$

$\therefore l = r\theta$

$= 3 \times \frac{\pi}{6}$

$= \frac{\pi}{2} \text{ cm}$

- 2 -

$$(f) \quad \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \tan^{-1}(1) = \frac{\pi}{6} + \frac{\pi}{4} = \frac{10\pi}{24} = \frac{5\pi}{12}$$

$$\boxed{k = \frac{5}{12}}$$

$$(g) \quad \int \frac{\ln 3x}{x} dx \quad \begin{array}{l} u = \ln 3x \\ \therefore du = \frac{3}{3x} dx = \frac{dx}{x} \end{array}$$

$$= \int \ln 3x \times \frac{dx}{x}$$

$$= \int u du = \frac{1}{2} u^2 + C$$

$$= \frac{1}{2} \ln^2 3x + C$$

$$(2) \quad (a) \quad (i) \quad \frac{d(x \cos x)}{dx} = x(-\sin x) + (1) \cos x$$

$$= \cos x - x \sin x$$

$$(ii) \quad \frac{d(\ln(\cos 5x))}{dx} = \frac{1}{\cos 5x} \times -5 \sin 5x$$

$$= -\frac{5 \sin 5x}{\cos 5x} = -5 \tan 5x$$

$$(iii) \quad \frac{d\left(\tan^{-1}\left(\frac{x}{3}\right)\right)}{dx} = \frac{1}{3} \times \frac{1}{1 + \left(\frac{x}{3}\right)^2} = \frac{1}{3} \times \frac{9}{9 + x^2}$$

$$= \frac{3}{9 + x^2}$$

$$(iv) \quad \frac{d(e^{\tanh x})}{dx} = \sec^2 x e^{\tanh x}$$

$$(v) \quad \frac{d\left([1 + \cos^{-1}(3x)]^3\right)}{dx} = 3[1 + \cos^{-1}(3x)]^2 \times \frac{-3}{\sqrt{1-9x^2}} = -\frac{9[1 + \cos^{-1}(3x)]^2}{\sqrt{1-9x^2}}$$

$$\begin{aligned}
 2(b) \quad (i) \quad & \int x^2 e^{2x^3+1} dx \\
 &= \frac{1}{6} \int (6x^2) e^{2x^3+1} dx \quad \left[\text{N.B. } \frac{d(2x^3+1)}{dx} = 6x^2 \right] \\
 &= \frac{1}{6} e^{2x^3+1} + C
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & \int \frac{5}{4+x^2} dx = \frac{5}{2} \int \frac{2}{4+x^2} dx \\
 &= \frac{5}{2} \tan^{-1}\left(\frac{x}{2}\right) + C
 \end{aligned}$$

$$(iii) \quad \int \frac{x^2}{x^3+7} dx = \frac{1}{3} \int \frac{3x^2}{x^3+7} dx = \frac{1}{3} \ln|x^3+7| + C$$

$$\begin{aligned}
 (iv) \quad & -\sin(\pi-x) = -\sin x \\
 \therefore \int -\sin x dx &= \cos x + C
 \end{aligned}$$

(3) (a) PRBN IG

$$\begin{aligned}
 3 \text{ letters} &= \binom{6}{3} = 20 & \frac{12}{20} &= \frac{3}{5} \\
 1 \text{ vowel} &= \binom{4}{2} \times \binom{2}{1} = 6 \times 2
 \end{aligned}$$

$$(b) (i) \quad 6 \text{ numbers} \Rightarrow 5! = 120$$

$$(ii) \quad 135 \quad 246$$

No even numbers together means alternating



$$\therefore 2! \times 3! = 2 \times 3 = 12 \text{ ways}$$

$$\therefore \text{Prob} = \frac{12}{120} = \frac{1}{10}$$

$$\therefore \text{At least 2} = 1 - \frac{1}{10} = \frac{9}{10}$$

-4-

(c) (i) $f(x) = x^2 + e^{-\frac{1}{2}x} - 5$ is continuous

$$f(-2) = 4 + e^{-1} - 5 = e^{-1} - 1 > 0 \quad (\because e > 3)$$

$$f(-1) = 1 + e^{\frac{1}{2}} - 5 = e^{\frac{1}{2}} - 4 < 0$$

$\therefore f(-2), f(-1) < 0$ and with f continuous

$\exists c$ s.t. $f(c) = 0, -2 < c < -1$

(ii) $x_0 = -2$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \left[\begin{array}{l} f(-2) = e^{-1} \\ f'(x) = 2x - \frac{1}{2}e^{-\frac{1}{2}x} \\ f'(-2) = -4 - \frac{1}{2}e \\ = \frac{-8 - e}{2} = -\frac{e+8}{2} \end{array} \right]$$

$$\therefore x_1 = -2 - \frac{e^{-1}}{-\left(\frac{e+8}{2}\right)}$$

$$= -2 + \frac{2(e^{-1})}{e+8} = \frac{-2(e+8) + 2(e^{-1})}{e+8}$$

$$= \frac{-2e - 16 + 2e^{-1}}{e+8}$$

$$= -\frac{16}{e+8}$$

(d) (i) $\frac{1+x}{1-x} = \frac{-(1-x) + 2}{1-x}$

$$= -\frac{(1-x)}{1-x} + \frac{2}{1-x}$$

$$= -1 + \frac{2}{1-x}$$

(ii) $\int \frac{1+x}{1-x} dx = \int \left(-1 + \frac{2}{1-x}\right) dx$

$$= -x + -2\ln|1-x| + c$$

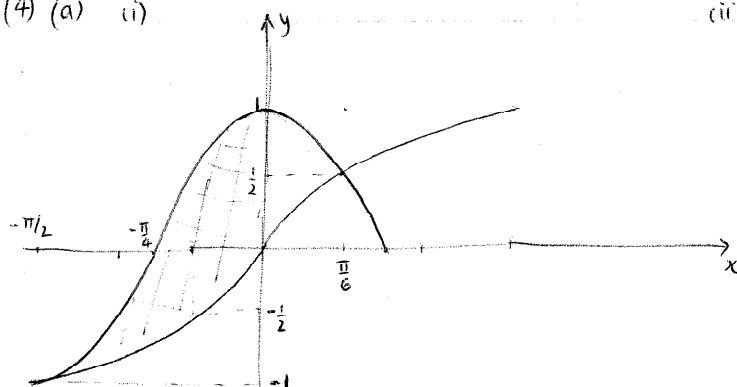
$$= -x - 2\ln|1-x| + c$$

$$3(e) \quad \sqrt{3} \tan \theta = 1$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = n\pi + \frac{\pi}{6}$$

(4) (a) (i)



$$(ii) \quad \sin\left(-\frac{\pi}{2}\right) = -1$$

$$\cos\left(2 \times -\frac{\pi}{2}\right) = \cos(-\pi) = -1$$

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

$$\cos\left(2 \times \frac{\pi}{6}\right) = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$(iii) \quad \text{Area} = \int_{-\pi/2}^{\pi/6} (\cos 2x - \sin x) dx = \left[\frac{1}{2} \sin 2x + \cos x \right]_{-\pi/2}^{\pi/6}$$

$$= \left(\frac{1}{2} \sin \frac{\pi}{3} + \cos \frac{\pi}{6} \right) - (0)$$

$$= \frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$$

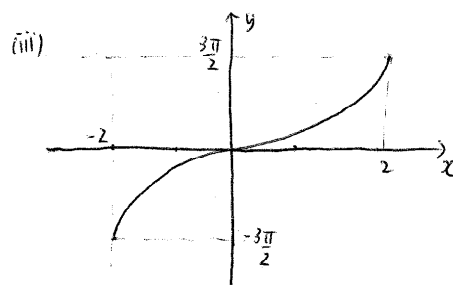
$$= \frac{3}{2} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$$

$$(b) \quad f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$$

$$(i) \quad f(2) = 3 \sin^{-1}(1) = 3 \times \frac{\pi}{2} = \frac{3\pi}{2}$$

$$(ii) \quad -1 \leq \frac{x}{2} \leq 1 \Rightarrow -2 \leq x \leq 2$$

$$-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2} \Rightarrow -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$



$$4(c) \quad \tan \left[2 \tan^{-1} \left(-\frac{1}{2} \right) \right] \quad \left\| \quad -\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2} \right.$$

$$\text{let } \alpha = \tan^{-1} \left(-\frac{1}{2} \right)$$

$$\therefore \tan \alpha = -\frac{1}{2}$$

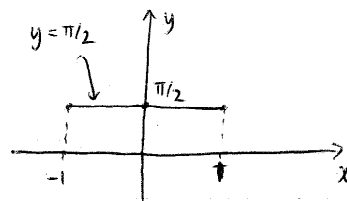
$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{2 \times \left(-\frac{1}{2} \right)}{1 - \left(-\frac{1}{2} \right)^2} = \frac{-1}{1 - \frac{1}{4}} = \frac{-1}{\frac{3}{4}} = -\frac{4}{3}$$

$$(d) \quad g(x) = \cos^{-1} x + \sin^{-1} x$$

$$(i) \quad g'(x) = \frac{-1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}} = 0$$

$$(ii) \quad \therefore g(x) = \text{constant for } -1 \leq x \leq 1$$

$$g(0) = \cos^{-1}(0) + \sin^{-1}(0) = \frac{\pi}{2} + 0 = \frac{\pi}{2}$$



$$(5) (a) \quad f(x) = x \sin^{-1}(x^2)$$

$$(i) \quad -1 \leq x^2 \leq 1 \Rightarrow -1 \leq x \leq 1$$

$$\Rightarrow -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$(ii) \quad f'(x) = \sin^{-1}(x^2) + x \left(\frac{2x}{\sqrt{1-x^4}} \right)$$

$$= \sin^{-1}(x^2) + \frac{2x^2}{\sqrt{1-x^4}}$$

= even function

(iii)

x	0 ⁻	0	0 ⁺
f'(x)	+	0	+

$\therefore$ HPOI

$\nearrow$
 $\frac{+}{0}$
 $\searrow$

$$\leftarrow \text{as } f'\left(-\frac{1}{2}\right) = f'\left(\frac{1}{2}\right)$$

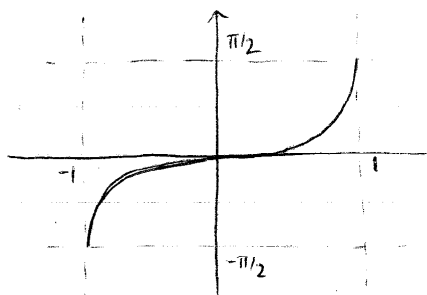
$$(iv) \quad f'(1) = \sin^{-1} 1 + \frac{2}{\sqrt{0}}$$

$\therefore$ vertical tangent

[OR: $\sin^{-1}(x^2)$ has a double root at $x=0$ $\therefore$ even

$\therefore x \sin^{-1}(x^2)$ has a triple root $\therefore$ HPOI]

5 (a) (v) $f(x)$ is an odd function



(b) (i) $x^2 + 2x - 3 = (x^2 + 2x + 1) - 4$
 $= (x+1)^2 - 4$

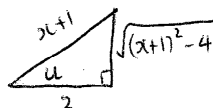
$$\therefore \int \frac{x}{\sqrt{x^2 + 2x - 3}} dx = \int \frac{x dx}{\sqrt{(x+1)^2 - 4}}$$

(ii) $2 \sec u = x+1$

$$\therefore 2 \sec u \tan u du = dx$$

$$\tan^2 u + 1 = \sec^2 u$$

$$[4 \sec^2 u - 4 = 4(\sec^2 u - 1) = 4 \tan^2 u]$$



$$\therefore \tan u = \frac{\sqrt{x^2 + 2x - 3}}{2}$$

$$\int \frac{x dx}{\sqrt{(x+1)^2 - 4}}$$

$$= \int \frac{(2 \sec u - 1) 2 \sec u \tan u du}{\sqrt{4 \sec^2 u - 4}}$$

$$= \int \frac{2(2 \sec u - 1) \sec u \tan u du}{2 \tan u}$$

$$= \int (2 \sec^2 u - \sec u) du$$

$$= 2 \tan u - \int \sec u du$$

$$= \sqrt{x^2 + 2x - 3} - \int \sec u du$$

could be done reversing
the substitution
BUT worth more than
2 marks!

not an Ext 1
integral

$$= \sqrt{x^2 + 2x - 3} - \ln |\sec u + \tan u| + C$$

$$= \sqrt{x^2 + 2x - 3} - \ln \left| \frac{x+1}{2} + \frac{\sqrt{x^2 + 2x - 3}}{2} \right| + C$$

$$= \sqrt{x^2 + 2x - 3} - \ln |x+1 + \sqrt{x^2 + 2x - 3}| + K$$

$$(5) \quad (c) \quad y = \frac{1}{\sqrt{1+x^2}}$$

$$\begin{aligned} V &= \pi \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} y^2 dx \\ &= \pi \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{1+x^2} dx \\ &= \pi \left[\tan^{-1}(x) \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \\ &= \pi \left[\tan^{-1}(\sqrt{3}) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \right] \\ &= \pi \left[\frac{\pi}{3} - \frac{\pi}{6} \right] \\ &= \pi \left[\frac{\pi}{6} \right] \\ &= \frac{\pi^2}{6} \text{ c.u.} \end{aligned}$$

$$(6) \quad a) \quad \int_0^{\sqrt{2}} x^3 \sqrt{x^2+1} dx \quad \equiv \frac{1}{2} \int_0^{\sqrt{2}} \sqrt{x^2+1} (2x dx)$$

$$\begin{aligned} u &= x^2+1 & \therefore du &= 2x dx \\ x=0 &\Rightarrow u=1 \\ x=\sqrt{2} &\Rightarrow u=3 \end{aligned} \quad \left| \begin{aligned} &= \frac{1}{2} \int_1^3 u^{\frac{1}{2}} du \\ &= \frac{1}{2} \times \frac{2}{\frac{3}{2}} u^{\frac{3}{2}} \Big|_1^3 \\ &= \frac{3}{8} \left[3^{\frac{3}{2}} - 1 \right] \\ &= \frac{3}{8} \left[\sqrt{27} - 1 \right] \\ &= \frac{3}{8} \left[3\sqrt{3} - 1 \right] \end{aligned} \right.$$

$$27 \times 3 = 81$$

6(b) $\cos 3x = \sin x$

(i) $\cos\left(\frac{\pi}{2} - A\right)$
 $= \cos\frac{\pi}{2} \cos A + \sin\frac{\pi}{2} \sin A$
 $= 0 \times \sin A + 1 \times \sin A$
 $= \sin A$

(ii) $\cos 3x = \sin x$
 $\Rightarrow \cos 3x = \cos\left(\frac{\pi}{2} - x\right)$

$\therefore 3x = 2n\pi \pm \left(\frac{\pi}{2} - x\right)$

$$\begin{array}{l} 3x = 2n\pi + \frac{\pi}{2} - x \\ 4x = (4n+1)\pi \\ x = \left(\frac{4n+1}{4}\right)\pi \end{array} \quad \left| \quad \begin{array}{l} 3x = 2n\pi - \frac{\pi}{2} + x \\ 2x = \frac{(4n-1)\pi}{2} \\ x = \frac{(4n-1)\pi}{4} \end{array} \right.$$

$\therefore x = \left(\frac{4n \pm 1}{4}\right)\pi$

(c) (i) $\frac{d(\tan^3 x)}{dx} = 3\tan^2 x \cdot \sec^2 x$
 $= 3(\sec^2 x - 1) \sec^2 x$
 $= 3\sec^4 x - 3\sec^2 x$

(ii) $\therefore \tan^3 x = \int (3\sec^4 x - 3\sec^2 x) dx$

$\therefore \tan^3 x \Big|_0^{\pi/4} = 3 \int_0^{\pi/4} \sec^4 x dx - 3 \int_0^{\pi/4} \sec^2 x dx$

$\therefore 3 \int_0^{\pi/4} \sec^4 x dx = \left[\tan^3 \frac{\pi}{4} - \tan^3(0) \right] + 3 \tan x \Big|_0^{\pi/4}$
 $= (1-0) + 3(1-0)$
 $= 4$

$\therefore \int_0^{\pi/4} \sec^4 x dx = \frac{4}{3}$

$$\cos 2A = 1 - 2\sin^2 A$$

$$(6)(d) \int_0^{\pi/6} \sin^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\pi/6} 2\sin^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\pi/6} (1 - \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/6}$$

$$= \frac{1}{2} \left[\pi/6 - \frac{1}{2} \sin \pi/3 \right]$$

$$= \pi/12 - \frac{1}{4} \times \frac{\sqrt{3}}{2}$$

$$= \pi/12 - \frac{\sqrt{3}}{8}$$

$$= \frac{2\pi - 3\sqrt{3}}{24}$$